

Dark Store Last-Mile Dispatch Playbook

Sample operations document • For DispatchDesk RAG corpus • Illustrative thresholds only, not a real company's policy

1. The 10-Minute Promise: Where the Time Goes

A quick-commerce delivery has three stages, and delay almost always lives in one of them. **Pick-pack** runs from order placed to packed and ready (target about 3 minutes). **Rider wait** is the time a packed order sits before a rider picks it up (target about 1.5 minutes). **Ride** is pickup to doorstep (target about 5 minutes within a 2 km radius). Average delivery time is the sum of the three, so always diagnose by comparing the stage breakdown against a normal evening rather than looking at the SLA percentage alone.

An SLA percentage falls off a cliff, not a slope: when the average delivery time drifts from roughly 8 minutes to 12 minutes, the share of orders inside 10 minutes can fall from around 90% to under 50%.

2. Diagnosing Delays

Rider wait is the dominant increase when packed orders are stacking up and few riders are free. Check the ratio of pending orders to available riders: above roughly 2 pending orders per available rider, waits climb quickly. Also check riders online versus a normal evening; riders dropping off in bad weather while orders rise is the classic pattern.

Pick-pack is the dominant increase when orders sit in the picking state, pickers are short, or a popular item is out of stock or mis-slotted. More riders will not fix this.

Ride time is the dominant increase when it is raining, there is a traffic event, or a large share of orders are in the outer zone. Ride time rises in wet weather even with no change in rider count.

3. Order Batching Rules

Batching puts two orders on one rider to relieve rider shortage. It is a tool for surges, not a default, because each extra order adds wait for the first customer. A batch is allowed only when **all** of these hold: (a) at most 2 orders per rider (3 only with explicit manager approval during a declared surge); (b) both drops are in the same or adjacent zones, and the added detour is no more than about 0.8 km or 2 minutes; (c) combined item count is 15 or fewer, so it fits one rider bag; (d) neither order contains frozen or ice-cream items, which are cold-chain items and must be delivered single-drop; (e) the older order has not already passed 8 minutes in the queue, since a batch would push it further. Always dispatch the oldest waiting order first.

4. Rain and Surge Playbook

In heavy rain, expect riders to drop offline, ride times to rise by about 30 to 40 percent, and order volume to jump. Recommended levers, in order: (1) call in standby riders (typically 20 minutes away); (2) apply policy-compliant batching for eligible orders; (3) activate the pre-approved surge incentive, capped at the store's approved per-order amount, with anything above the cap needing regional ops sign-off; (4) temporarily shrink the serviceability radius from 3 km to 2 km, or pause the outermost zone; (5) show customers an honest, longer ETA instead of holding the 10-minute promise. Do not treat a missed SLA in heavy rain as a rider performance problem.

5. Rider Safety and Working Hours Policy

Riders must never be asked, pressured, or incentivized to break traffic rules, speed, or ride in unsafe conditions to hit a delivery time. No message, incentive, or penalty may reward unsafe riding. Riders are never penalized

(pay deductions, warnings, or rating impact) for delays caused by weather, traffic, or safety-driven decisions. In heavy rain or flooding, a rider may decline or pause work without penalty.

Maximum shift: 9 hours on shift, including breaks. **Mandatory break:** at least 15 minutes after every 4 hours without a break. A rider who has reached either limit is not to be dispatched or scheduled further, regardless of demand. Overtime beyond these limits is not available to a store manager at any price; if capacity is short, use standby riders, rebalance zones, or request extra riders from regional ops.

6. Customer Communication and ETAs

Show customers an ETA based on current conditions, and never promise a time the data does not support. Send a proactive delay notice when an order passes 8 minutes in the queue. Any ETA is an estimate: give a range, not a guarantee. Customer notifications and compensation credits are drafted for the manager to approve before sending.

7. Quick Reference: Symptom, Cause, First Action

Symptom	Likely cause	First action	Escalate to regional ops when
High rider wait, pending/available above 2	Too few riders online or free	Dispatch oldest order first; call standby rider; batch eligible orders	No standby available and wait stays high for 15+ minutes
High pick-pack time, orders stuck in picking	Picker shortage, stock-out, mis-slotting	Reassign a picker; check top-10 items for stock	Stock-out of a high-volume item
High ride time, rain flag on	Wet roads, slower riding	Show longer ETA; shrink radius; apply surge incentive within cap	Flooding, or incentive above cap needed
Rider near shift or break limit	Long continuous work	Stop dispatching to that rider; arrange break or relief	Relief cannot be arranged
Frozen item in a waiting order	Cold-chain constraint	Single-drop only; dispatch promptly	N/A
Many outer-zone (Z-C) orders waiting	Long rides consuming capacity	Pair only with adjacent zones; consider pausing outer zone	Sustained for 30+ minutes